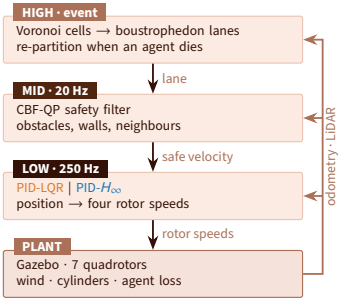
Fault-Tolerant Swarm Quadrotor Control for Non-Convex Geodetic Mapping

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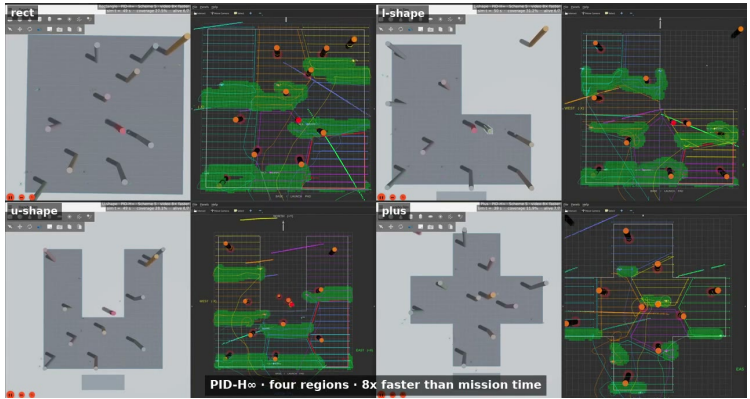
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github.com/izmaherdian/swarm-quadrotor-mapping

Seven quadrotors, four regions, every stressor at once



PID-H_∞ on rect, L, U and plus · 8× faster than mission time · Gazebo left, RViz map right · Click the frame to play

Seven of eight missions finished after losing 28.6% of the fleet

- The problem — non-convex area, wind, unmapped obstacles, agent loss
- Three layers — who sweeps where, a safety filter, two controllers
- Eight recorded missions, including the one that crashed
- What we would not yet claim

THE NUMBERS BEHIND THAT SENTENCE

96.9–99.0% coverage · PID-H_∞ finished 11–15% sooner than PID-LQR, at 3.9–5.6× the torque effort · one mission ended in an obstacle strike.

1

The problem

Why one drone is not enough — and why a swarm is not automatically better

The question is not whether a swarm is faster, but whether it still finishes

- Non-convex region — notches and arms break a naive raster
- Wind — mean 3.0 m/s, gusts to 12.0 m/s
- Nine cylinders the swarm has never seen, plus two moving
- Two quadrotors killed mid-mission, at 18% and 32% coverage

$$T_{\text{cover}} \sim \frac{A}{N \cdot 2 r_s \cdot \bar{v}} + T_{\text{coord}}(N)$$

ALL FOUR AT ONCE

Not one at a time. Sweep time falls with agent count; coordination cost rises with it.

Four mechanisms, and a campaign honest enough to include a crash

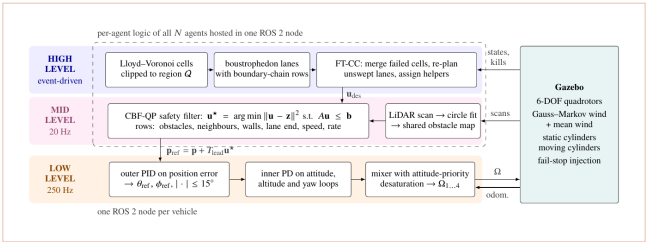
- Lanes that follow the cell boundary, so tapered corners get swept
- Lane-level reallocation when an agent dies — merge, re-lane, hand over
- A safety filter whose speed limit comes from the real tilt limit
- PID-LQR vs PID-H_∞ on an identical structure — only the gains differ
- Eight logged missions, an auditable number pipeline, one failure

2

The system as blocks

What enters each layer, what it computes, what it hands down

Three nested loops, each one handing a set-point to the next

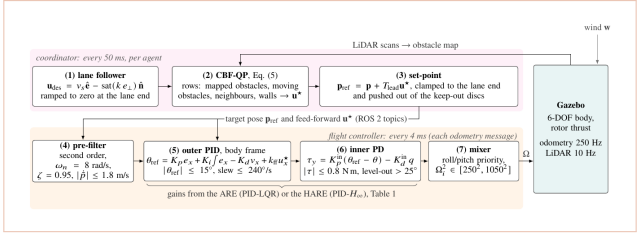


HIGH · event-driven
in region polygon, agent poses, agent deaths
out a lane to follow

MID · 20 Hz
in desired velocity, LiDAR map, neighbours
out a safe velocity

LOW · 250 Hz
in position set-point, state
out four rotor speeds

We derive the stack backwards: what the rotors allow decides what the planner may ask



THE CHAIN, IN THREE STEPS

Low tilt limit → velocity plant. **Mid** plant → stopping distance → $\varphi(h)$ → one row per hazard. **High** $\varphi(h)$ and sensor footprint → keep-out radius and lane pitch. Nothing upstairs is tuned by hand.

3 Low level

From rotor limits to a one-line velocity plant

Low level: a position in metres goes in, four rotor speeds come out

WHAT GOES IN

set-point
p* [m]
from the mid level

measured state
p [m], **v** [m/s]
η [rad], ω [rad/s]

WHAT IS COMPUTED

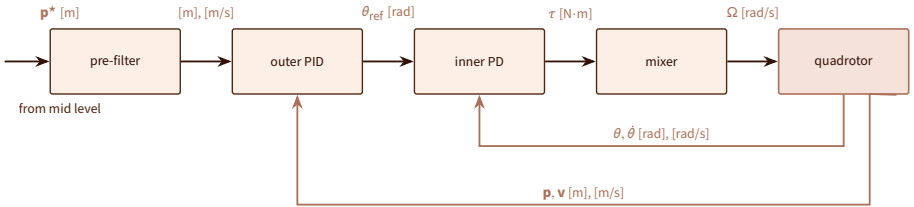
pre-filter, outer PID, inner PD, mixer — a fixed chain, no optimisation at run time

every 4 ms

WHAT COMES OUT

four rotor speeds
$\Omega_{1..4}$ [rad/s]
bounded to
[250, 1050]

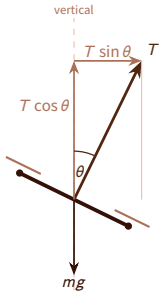
effect: thrust and torque



the whole chain runs 250 times per second

Step 1 — the only horizontal force a quadrotor has is tilted thrust

- Four rotors give one force, along the body z axis
- Hover needs $T\cos\theta = mg$; what is left sideways is $T\sin\theta$
- Divide the two and the mass drops out — no mass, no thrust, just the tilt angle
- Cap the tilt for stable mapping flight and the acceleration is capped with it



$$\theta \le 15^\circ \implies a_{\max} = g\tan\theta_{\max}$$

WHERE 15° COMES FROM

A design choice, not physics: the mapping sensor needs a steady attitude. It caps the attitude *reference*, which is why measured tilt exceeds it at lane-end pivots (Q11).

$2.631\ \mathrm{m/s^2}$

is every bit of horizontal acceleration this vehicle has

$a_{\max} = g \tan 15°$. About a quarter of g. Each of the next nine slides uses this number; nothing downstream hardcodes physics.

Step 2 — twelve coupled states become four simple chains

- A quadrotor carries twelve states: position [m], velocity [m/s], attitude [rad] and body rates [rad/s]
- Nonlinear, because rotation couples all of them: you tilt to move, and the thrust direction moves with you
- Linearise about hover with small angles and those couplings vanish to first order
- What is left is four independent subsystems

X → pitch	$x \leftarrow \theta \leftarrow \tau_y$
Y → roll	$y \leftarrow \phi \leftarrow \tau_x$
Z	$z \leftarrow$ collective thrust
yaw	$\psi \leftarrow \tau_z$

$$\underbrace{\mathbf{p}, \mathbf{v}, \eta, \omega}_{\text{12 states, nonlinear}} \xrightarrow{\text{hover}} \underbrace{4 \text{ chains}}_{\text{2 states each}}$$

WHERE THIS STOPS BEING TRUE

The linearisation is why the 15° cap matters. The upsets we measured, 82.5° and 155.9°, lie outside its validity — so we report them as events, not as tracking error.

Step 3 — augment with the input, and the integral term falls out

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \; B = \begin{bmatrix} 0 \\ b \end{bmatrix}, \; C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$b = g$ (outer X), $-g$ (outer Y), $1/I_{yy}$, $1/I_{xx}$ (inner)

$$A_{\text{aug}} = \begin{bmatrix} A & B \\ 0 & 0 \end{bmatrix}, \quad B_{\text{aug}} = \begin{bmatrix} 0 \\ I \end{bmatrix}$$

- The augmented state carries the *input* itself, so the design variable becomes $\dot{u}$ — feed back a rate and you have already integrated once
- LQR solves the Riccati equation on $(A_{\text{aug}}, B_{\text{aug}}, Q, R)$; H_∞ solves the HARE with $\gamma = 1.3\gamma_{\min}$ and a Hurwitz check
- K_a is then projected onto the PID structure through Γ^+, giving K_p, K_i, K_d

SO K_i IS NOT A TUNING KNOB

It is a consequence of designing on $\dot{u}$. The two controllers differ only in Q, R and γ — never in structure.

Step 4 — same augmented plant, two Riccati equations

PID-LQR — optimal for the model you wrote

$$J = \int_0^\infty \left(\tilde{x}^\top \tilde{Q} \tilde{x} + \dot{u}^\top R\, \dot{u}\right) \mathrm{d}t$$

$$\tilde{A}^\top P + P\tilde{A} - P\tilde{B} R^{-1} \tilde{B}^\top P + \tilde{Q} = 0$$

$$K_a = R^{-1} \tilde{B}^\top P$$

PID-H_∞ — optimal against the worst disturbance

$$\text{find } K: \quad \|T_{w \to z}\|_\infty < \gamma$$

$$\tilde{A}^\top P + P\tilde{A} + P\left(\gamma^{-2} B_w B_w^\top - B_u R^{-1} B_u^\top\right) P + \tilde{Q} = 0$$

$K_a = R^{-1} B_u^\top P$, with $\gamma = 1.3\, \gamma_{\min}$ by bisection

ONE SOLVER, TWO PROBLEMS

The HARE is obtained from the same Riccati routine by stacking $\tilde{B} = [B_u \;\; B_w]$ and setting $\tilde{R} = \mathrm{diag}(R, \; -\gamma^2 I)$ — the negative block is what turns “cheapest” into “safest against the worst case”. Every solution is Hurwitz-checked on $\tilde{A} - B_u K_a$ before it is accepted.

Step 5 — project the state feedback onto a PID, and that is what flies

$$\Gamma = \begin{bmatrix} C^\top & (CA)^\top & (CA^2)^\top \\ 0 & (CB)^\top & (CAB)^\top \end{bmatrix}, \qquad \hat{K} = \Gamma^{+} K_a^\top$$

$$K_d = \hat{k}_3 \left(I + CB\,\hat{k}_3\right)^{-1}$$

$$K_p = (I - K_d\,CB)\hat{k}_2, \qquad K_i = (I - K_d\,CB)\hat{k}_1$$

- Γ^{+} is a pseudo-inverse: the optimal state feedback is *projected* onto what a PID can express
- So the gains are not fitted — they are the closest PID to an optimal controller
- What runs on the vehicle is still an ordinary PID, which is why the two designs are swappable at run time

THE LAW THAT ACTUALLY EXECUTES, 250 TIMES A SECOND

$\theta_{\text{ref}} = K_p\,e + K_i \int e\,\mathrm{d}t - K_d\,v + k_{ff}\,v_{\text{cmd}}$, saturated at $|\theta_{\text{ref}}| \le 15°$ — and the integrator is gated at $|e| < 0.25\,\text{m}$ for LQR, always active for H_∞. That one asymmetry returns in the results.

Step 6 — the outer PID plus a lead reference collapse into a velocity plant

- The outer loop forms an attitude reference from position error, measured velocity and a velocity feedforward; for small angles $a \approx g\,\theta_{\text{ref}}$
- The coordinator never sends a velocity. It sends a position that *leads* the agent by $T_{\text{lead}}\,v_{\text{cmd}}$, so the position error *is* the command

$$\theta_{\text{ref}} = K_p e + i - K_d v + k_{ff}\,v_{\text{cmd}}, \qquad e = T_{\text{lead}}\,v_{\text{cmd}}$$

$$a = \underbrace{g\,(K_p T_{\text{lead}} + k_{ff})}_{\text{gain on command}}\,v_{\text{cmd}} \;-\; \underbrace{g K_d}_{k_v}\,v$$

CHOOSE T_{lead} SO THE DC GAIN IS EXACTLY ONE

$g(K_p T_{\text{lead}} + k_{ff}) = g K_d$ gives $T_{\text{lead}} = (K_d - k_{ff})/K_p$: a commanded speed is the speed actually held, and the closed loop is one first-order equation.

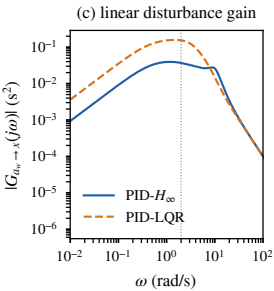
Step 7 — four numbers, and they come from the gains, not from tuning

$$\dot{v} = \mathrm{sat}_{\pm a_{\max}}\big(k_v\,(v_{\mathrm{cmd}} - v)\big)$$

$k_v = gK_d$	`3.867`	s^{-1}
$a_{\max} = g\tan\theta_{\max}$	`2.631`	$\mathrm{m/s}^2$
$T_{\mathrm{lead}} = (K_d - k_{ff})/K_p$	`0.323`	s
$v_c = a_{\max}/k_v$	`0.680`	m/s

- v_c is where saturation stops binding — below it the response is exponential, above it acceleration-limited
- Change a gain or the tilt cap and all four move together
- The same two designs, LQR and H_∞, differ only here: $\|G\|_\infty$ `0.158` → `0.040` s^2 (4.0×), K_i 3.9× larger, phase margin 37.4° → 25.2°

What Q, R and γ actually bought, before any flight



Disturbance-to-position gain · dotted line is the wind corner frequency

- Worst-case gain from disturbance to position: $\|G\|_\infty$ `0.158` → `0.040` s^2, a 4.0× reduction
- Predicted deviation under this wind falls 3.7× (σ_x `0.115` → `0.031`)
- Bought with bandwidth: crossover 4.09 → 9.08 rad/s
- Paid for in margin: phase 37.4° → 25.2°, gain 11.5 → 9.0 dB

A PREDICTION, NOT A RESULT

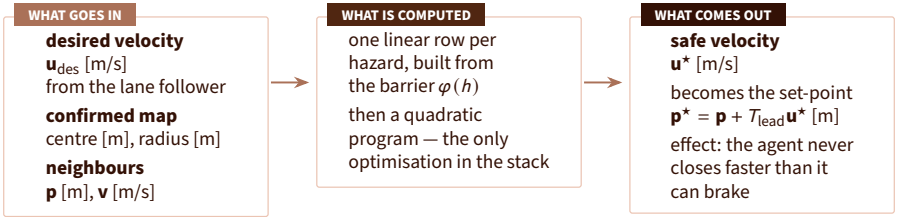
This is the linear model talking. Whether the flown data agrees — and whether it agrees for *this* reason — is the tracking slide in the results.

4

Mid level

From the velocity plant to one linear row per hazard

Mid level: a wish in metres per second goes in, a safe one comes out



What is optimised: the *deviation from what the planner asked*, $\|\mathbf{u} - \mathbf{u}_{\text{des}}\|^2$ — never the safety rows, which are hard constraints.

Runs 20 times per second, per agent. The output is the box marked “set-point” on the previous section — this is where the two layers meet.

Step 8 — integrate the plant and you get an exact stopping distance

- Command zero and the plant decelerates at $\min(k_v v,\ a_{\max})$
- Fast: acceleration-limited, a parabola. Slow: exponential
- The two regimes meet exactly at v_c

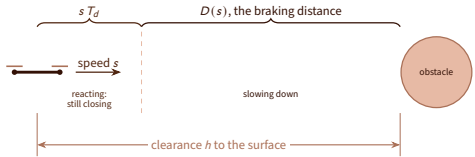
$$D(s) = \begin{cases} \dfrac{s}{k_v}, & s \le v_c \\ \dfrac{s^2 + v_c^2}{2a_{\max}}, & s > v_c \end{cases}$$

THIS IS EXACT, NOT A BOUND

$D(s)$ is the integral of the identified plant, so the safety argument below inherits whatever fidelity the identification has — no extra conservatism, and no optimism either.

Step 9 — add dead time, and safety becomes one inequality

- The agent reacts $T_d = 0.40$ s late: sensing, the QP, the set-point hand-off, actuator lag
- During that window it keeps closing at speed s; only afterwards does $D(s)$ begin, so the clearance h must cover both



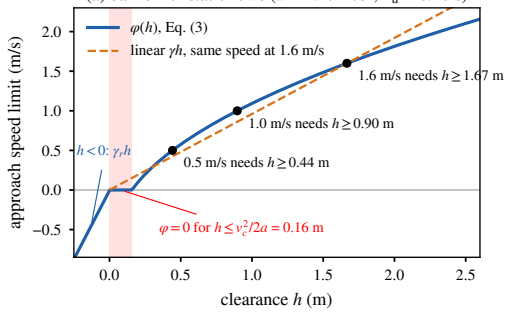
READ IT AS A SPEED LIMIT, NOT A DISTANCE CHECK

Linear in h, quadratic in s. Solving for s turns a safety requirement into something the controller can act on.

Step 10 — solve for the speed and the barrier $\varphi(h)$ falls out

$$\varphi(h) = \max\left(0, -aT_d + \sqrt{a^2T_d^2 + 2ah - v_c^2}\right), \quad \varphi(h) = \gamma_r h \ \text{ for } h < 0$$

- The fastest approach that can still be stopped — an extended class-$\mathcal{K}$ function with units of speed
- $\varphi = 0$ for $h \le v_c^2/2a = 0.16$ m: at that clearance the vehicle must already be stopped
- Below zero it goes negative and *commands* separation ($\gamma_r = 3.5\,\mathrm{s}^{-1}$)
- A linear γh matched at 1.6 m/s still permits approach inside 0.16 m — it fails exactly where it matters



1.67 m

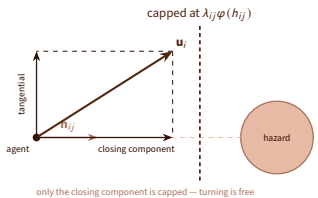
of clearance before 1.6 m/s is an allowed approach speed

1.0 m/s needs 0.90 m · 0.5 m/s needs 0.44 m · at 0.16 m the admissible speed is zero. Read off φ, with no parameter left to choose.

Step 11 — one hazard, one linear row on the commanded velocity

$$\mathbf{n}_{ij}^{\top}\mathbf{u}_i \;\le\; \lambda_{ij}\,\varphi(h_{ij}), \qquad \lambda_{ij} + \lambda_{ji} = 1$$

- $\mathbf{n}_{ij}$ points from agent to hazard; the row caps only the *closing* component, never the tangential one
- Static cylinders, moving cylinders, arena walls and neighbours all produce rows of this one shape
- Agent pairs split the burden reciprocally — exact only from one snapshot, so all seven are solved together
- Hard separation 0.70 m, soft 1.20 m



Step 12 — stack the rows and the safe velocity is a projection

$$\mathbf{u}^\star = \arg\min_{\mathbf{u}} \ \|\mathbf{u} - \mathbf{u}_{\text{des}}\|^2 \quad \text{s.t.} \quad A\mathbf{u} \le b$$

- The state machine proposes $\mathbf{u}_{\text{des}}$; the QP returns the nearest admissible velocity
- Two variables only, so it is solved *exactly* by active-set enumeration — $138\,\mu$s at 24 rows, no solver dependency
- Validated on 3000 random instances: SLSQP failed 320, this none
- Deadlock recovery touches only the objective, never A or b — liveness is never traded against safety

MEASURED OVER EIGHT MISSIONS

19,500–25,900 solves each. Tier 1–2 fallbacks on 2.91–4.94%; tier 3 never in a completed run.

5

High level

From $\varphi(h)$ and the sensor footprint to the lane grid

High level: a polygon goes in, a lane to fly comes out

WHAT GOES IN

region
polygon, [m^2]

agent poses
$\mathbf{p}_{1..7}$ [m]

coverage grid
0.30 m cells

events
an agent has died

→

WHAT IS COMPUTED

clip, partition, Lloyd, Hungarian assignment

boustrophedon lanes at $\leq$ 1.45 m pitch

keep-out routing at 1.30 m; on a loss, merge and re-lane

→

WHAT COMES OUT

one lane
two endpoints [m]

and a carrot on it, giving $\mathbf{u}_{des}$ [m/s]

effect: every cell gets swept, and a dead agent's cell still gets swept

No clock: this layer only runs when a lane ends, an obstacle is confirmed, or an agent dies.

Its output is the leftmost box of the mid level — the chain closes, and every limit it applies came from below.

Step 13 — the planner's keep-out radius is read off the barrier, not chosen

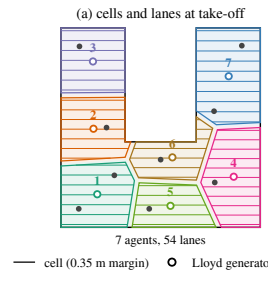
- Physical need: cylinder 0.40 + airframe 0.22 + margin 0.25 = `0.87` m
- But a detour is an *arc*, and a chord cuts inside it. Capping the chord at 70° gives $1.30 \cos 35° = $ `1.065` m — still clear of `1.10` m, the proved minimum
- So the planner routes at `1.30` m, and the string-pulling pass may only shorten a path if it can prove $\geq$ `1.10` m
- At `1.30` m with a `0.95` m sensor, the ring around each cylinder is still swept — the detour costs no coverage

PLANNER VERSUS FILTER

The planner keeps the nominal path clear of what is *known*. The QP handles what was not known yet. Between a LiDAR discovery and the replan, the QP is the only thing keeping the agent safe.

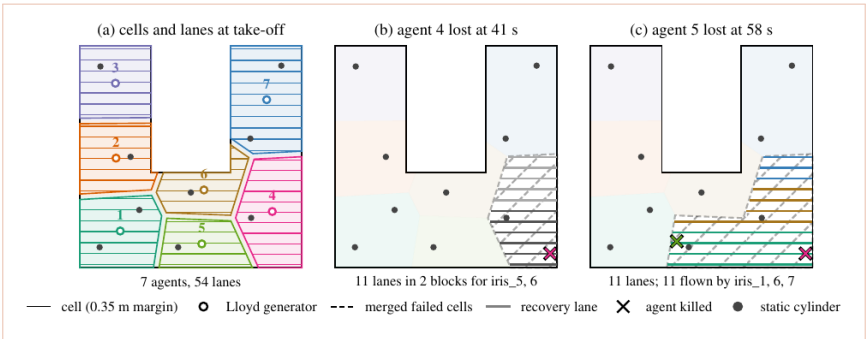
Step 14 — the sensor sets the lane pitch, the region sets the cells

- Sensor radius `0.95` m → `1.90` m swath; lanes at ≤ `1.45` m leave ≥ `0.45` m overlap, so heading error opens no gaps
- Cells: each agent takes the points nearest to it, generators move to the centroid, repeat — Lloyd, 25 passes, then Hungarian assignment on transit cost
- Every cell is clipped to the region polygon first and shrunk `0.35` m, so nothing assumes a rectangle
- First and last lanes follow the cell's own boundary chain — a fixed-pitch raster misses the wedges at tapered corners



Take-off: 7 agents, 54 lanes

Step 15 — when an agent dies, the same geometry is simply re-run



(a) 7 agents, 54 lanes · (b) agent 4 lost at 41 s · (c) agent 5 lost at 58 s

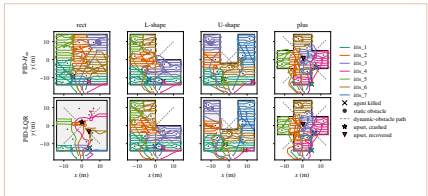
6

Experiment

Every stressor at once, eight recorded missions

Every stressor applied at once, to seven quadrotors

- Seven 3DR Iris in Gazebo + ROS 2 · cruise 2.0 m · sweep 1.6 m/s
- Wind: Gauss–Markov, $\sigma = 2.5$ m/s, plus a constant -2 m/s per axis
  - recorded mean (-2.34, -1.95) m/s, peaks 12.0 m/s
- Two moving cylinders oscillating diagonally at up to 2.5 m/s
- Two agents killed fail-stop, at 18% and 32% coverage



Flown trajectories — PID-H_∞ (top), PID-LQR (bottom)

Every number here traces to a CSV, or it is not shown

- Four regions × two controllers, recorded 11–12 September 2026
- Each run carries a manifest: recording, log, git commit, who was killed
- One analysis script over 56 CSVs and 8 logs generates every figure
- $N = 1$ per configuration, one wind seed — single-run observations

WHY WE SAY THIS UP FRONT

Earlier tooling in this project silently synthesised telemetry when CSVs were missing. Missing data now prints n/a and is never substituted.

7

The missions

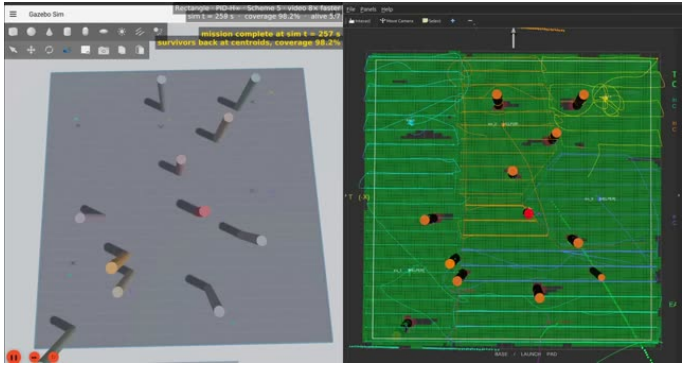
Four clips — one full mission, two mechanisms, one failure

7 of 8

missions finished, after losing two of seven quadrotors

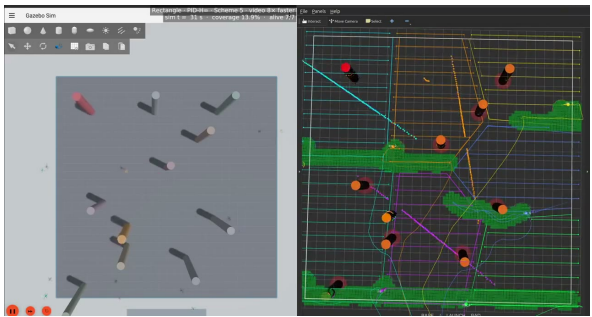
96.9–99.0% coverage under turbulence, unmapped static obstacles and moving obstacles. The eighth is four slides away, and we are going to watch it.

A full mission: partition, transit, sweep, two losses, return to centroid



rect, PID-H_∞ · 8× wall clock ≈ real mission time · 98.2% in 257 s, agents 4, 7 killed · Click the frame to play

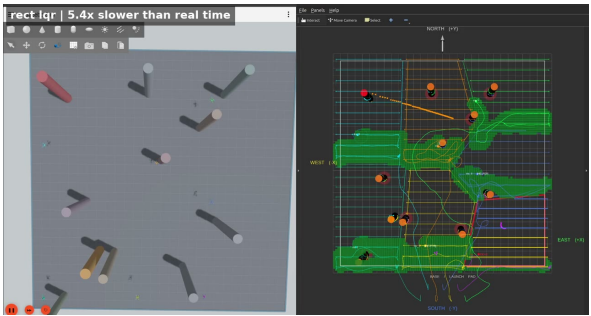
The swarm never pauses — survivors keep sweeping through a reallocation



rect, PID-H_∞ · 25 s of mission time around the first kill · Click the frame to play

- Agent 4 killed at 18% coverage
- Coverage inside its cell is erased
- The cell is merged and re-laned
- Helper groups this run: 7,6; 6,1,2

One mission ended in an obstacle strike, and we report it as such



Slowed 5.4× relative to real time · the only one of eight missions that did not finish · Click the frame to play

- Agent 2 reaches 0.06 m from a cylinder surface
- Inside the airframe envelope — 0.235 m body, 0.348 m rotors
- Tilt 155.9°, altitude 0.11 m, no recovery
- Stopped at 78.8 s; coverage reported n / a

8

Results

What finished, how tightly it flew, and what it cost

H_∞ finished every region; LQR finished three and struck a cylinder in the fourth

Region	Controller	Status	Cov.%	T (s)	lat. (cm)	$\bar{v}$ (m/s)	tilt_{95} / max (°)	effort	d_{obs} (m)	d_{v2v} (m)
rect	PID-H_∞	done	98.2	257	7.2	1.20	14.7 / 21.1	0.054	0.49	1.71
rect	**PID-LQR**	**abort @ 78.8 s**	**n/a**	**n/a**	**11.7**	**1.12**	**14.7 / 155.9**	**0.024**	**0.06**	**1.52**
L-shape	PID-H_∞	done	98.4	201	8.4	1.19	14.5 / 20.8	0.058	0.57	1.32
L-shape	PID-LQR	done	98.5	233	12.2	1.11	14.5 / 26.3	0.010	0.29	0.79
U-shape	PID-H_∞	done	98.5	202	8.3	1.21	14.8 / 20.7	0.044	0.63	1.77
U-shape	PID-LQR	done	96.9	237	11.4	1.07	14.2 / 16.6	0.011	0.42	1.73
plus	PID-H_∞	done	98.4	195	9.7	1.15	14.7 / 82.5	0.053	0.24	1.24
plus	PID-LQR	done	99.0	219	11.5	1.02	14.4 / 40.1	0.014	0.24	1.12

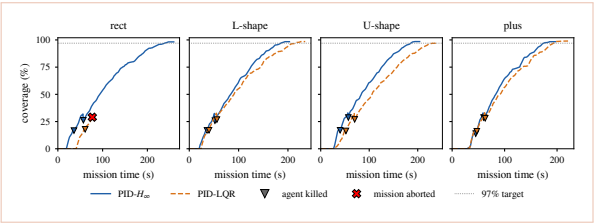
- LQR is not uniformly worse: best coverage of the campaign (99.0% in plus) and the lowest peak tilt (16.6° in U-shape)
- Sweep-phase tilt sits at the 15° reference cap in every run; the maxima come from lane-end pivots and upsets

0.06 m

was the clearance at which the one failure happened

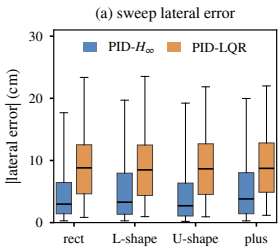
Inside the 0.235 m body half-width, let alone the 0.348 m rotor reach. Every upset in the campaign was preceded by 1–4 s of tier-2 QP solves bound by that same cylinder — the filter saw it coming and ran out of control authority.

Coverage goes backwards at every loss — by design



- Coverage inside a failed cell is erased, so the curve genuinely drops
- The cross marks the aborted rect / PID-LQR run at about 79 s
- PID-H_∞ (solid) reaches the 97% target sooner in every region it completed

Most of the H_∞ advantage is a rejected mean offset, not broadband attenuation



- RMS 11.4–12.2 → 7.2–9.7 cm — a 16–39% reduction
- But the spread is close: std 9.4–10.8 vs 7.0–9.6 cm
- The MEAN is what moved: LQR -7.0 to -5.7 cm, H_∞ 0.7–1.6 cm

TWO REAL EFFECTS, NOT ONE CLAIM

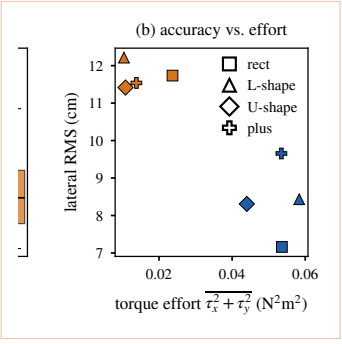
H_∞ does attenuate (4.0× lower $\|G\|_\infty$) and does have a 3.9× larger integral gain. Here the second dominates; saying only the first would overstate the method.

55–64%

of sweep samples had the LQR integrator open — against 84–87% for H_∞

The LQR position integrator is gated to $|e| < 0.25$ m, but the set-point leads by $T_{\text{lead}} u^\star \approx 0.4$ m during a sweep, so the gate is shut much of the time. That is the mechanism behind the lateral bias — and it is a tuning choice, not a property of LQR.

Holding the lane tighter costs torque, and the trade is one scatter



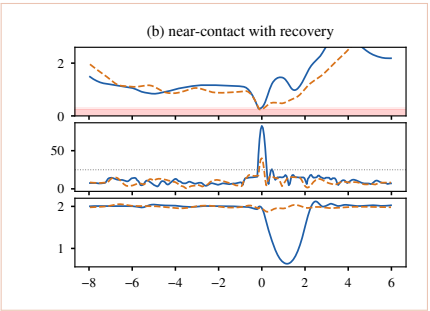
Lateral RMS against torque effort $\overline{\tau_x^2 + \tau_y^2}$ · marker shape is the region

- Every H_∞ run sits lower *and* further right: tighter lanes, more effort
- Torque effort 3.9–5.6× higher — `0.010–0.014` for LQR against `0.044–0.058` for H_∞
- Motor saturation stayed under 0.2% of samples for both: effort spent, not authority exhausted
- On a battery-limited platform this ratio, not the lateral error, may be what decides

NEITHER DESIGN WINS OUTRIGHT

H_∞ is tighter on the lane and harder on the airframe; LQR is gentler and looser.

No pair went below 0.70 m — which is not the same as no collisions



plus, iris_2 · PID-H_∞ (blue) vs PID-LQR (orange) · surface distance, tilt, altitude

- Closest agent pair of the campaign: 0.79 m vs a 0.70 m hard radius
- Static clearance — H_∞ 0.49 / 0.57 / 0.63 m, LQR 0.06 / 0.29 / 0.42 m
- Three missions passed within rotor reach without contact being provable

WHAT WE DO NOT CLAIM

Rotor guards are an assumption, not a simulated result, and inter-agent contact is never detected — only $d_{\min}$ is tracked.

What this work does not establish

- Single runs, one wind seed — $N = 1$ per configuration
- Perfect state and perfect moving-obstacle feedback
- Instant failure detection; geometric coverage only
- Rotor guards assumed, not simulated
- Centralised upper layers; no ablation, no re-implemented baseline

Seven of eight, under every stressor at once

- 96.9–99.0% coverage while losing two of seven quadrotors
- Where both finished, H_∞ held lanes 16–39% tighter and finished 11–15% sooner
- — at several times the torque effort, and with smaller margins
- One LQR mission struck a cylinder; three others came within rotor reach
- Next: several wind seeds, a guard-sized gust-aware barrier, distributed upper layers, flight tests

Thank you

Questions welcome

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Engineering Physics, Institut Teknologi Bandung

Code MIT, media CC BY 4.0 · all eight mission recordings are public
Acknowledgment: CITA ITB covered registration and publication fees



github.com/izmaherdian/swarm-quadrotor-mapping

B

Backup

Extra material — not part of the 20-minute talk

Questions we expect

Q1 Why a CBF-QP? The abstract said ORCA.
ORCA assumes an integrator that can change velocity instantly · slide 55

Q2 Why no PX4 SITL? Where is the estimator?
Ground-truth odometry, no PX4 in the loop; stated as a limitation · slide 56

Q3 $N = 1$. Is this statistically meaningful?
No — single-run observations, and the abstract says so · slide 57

Q4 Why did the LQR run crash? Is it fair?
Identical structure; the crash follows from the bias · slide 58

Q5 Is H_∞ better, or just a bigger integrator?
Both are real and we separate them · slide 59

Q6 Did any two drones collide?
Closest pair 0.79 m, but contact is never detected · slide 60

Q7 A barrier of -0.68 m is negative. Contact?
Cannot be confirmed or excluded; guards are an assumption · slide 61

Q8 What does the filter cost? Is it real-time?
138 μs per agent against a 50 ms budget · slide 62

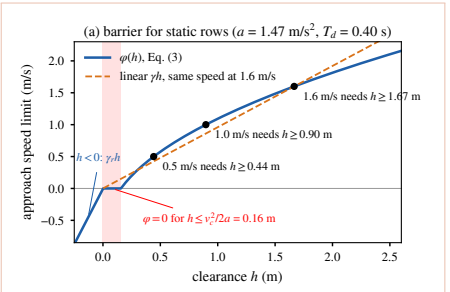
Q9 Does this scale past seven agents?
The QP does; the upper layers are centralised · slide 63

Q10 Why is coverage not 100%?
Geometric coverage; offline ceiling is 97.8–99.6% · slide 64

Q11 Tilt limit is 15° but the table shows 21–26°.
15° caps the reference, not the measured attitude · slide 65

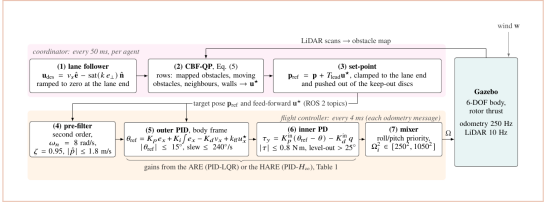
ORCA's guarantee assumes an actuator this vehicle does not have

- ORCA's half-plane is safe only if velocity can change instantly
- This quadrotor has 2.631 m/s^2 and 0.40 s of dead time
- So a half-plane can be geometrically feasible and actuator-infeasible
- A CBF row states the real limit: the fastest approach that can still stop



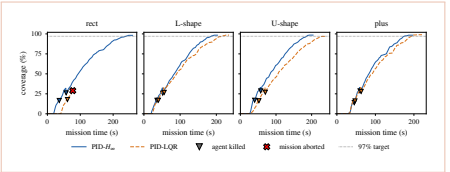
Removing the estimator is what keeps the controller comparison clean

- Controllers read ground-truth odometry and write rotor speeds directly
- No EKF2, commander, mixer or MAVLink — PX4 gives airframe parameters only
- Upside: the LQR–H_∞ difference is not confounded by estimator dynamics
- Cost: no claim about state noise, latency or GPS dropout. Listed as a limit



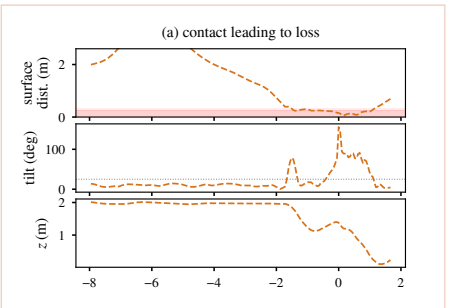
No — and we call them single-run observations in the abstract

- Eight missions, one wind seed. Not statistics
- Still readable: the mechanisms, and a lateral bias that appears in all four regions
- Not readable: ranking by a few percent of time, or crash probability from one crash
- What it would take: several seeds × repetitions, paired statistics



The crash is a consequence of the bias, not of an unfair setup

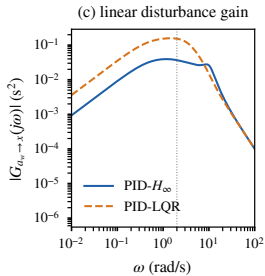
- Shared: cascade, every limit, planner, QP, scenario. Different: gains and the integrator gate
- The chain — -7.0 to -5.7 cm downwind bias → 0.06 m clearance → tilt 155.9° → no recovery
- An earlier upset in the same run recovered: iris_6, 50.8° at 0.55 m
- Reported as abort with coverage n / a — not scored partially, not dropped



rect, PID-LQR, iris_2 · surface distance, tilt, altitude

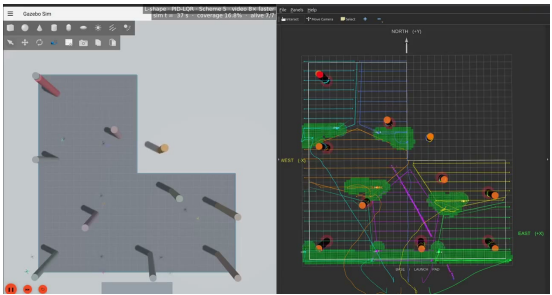
Both effects are real — and here the integrator policy dominates

- Design side: $\|G\|_\infty$ 0.158 → 0.040 s^2, a 4.0× reduction; predicted deviation falls 3.7×
- Flight side: the spread barely moved — the mean did
- H_∞ integral gain is 3.9× larger; the LQR gate is shut much of a sweep
- Measured open fraction: 55–64% vs 84–87%



Disturbance-to-position gain · dotted line is the wind corner frequency

No pair went below 0.70 m — which is a weaker statement than no collisions

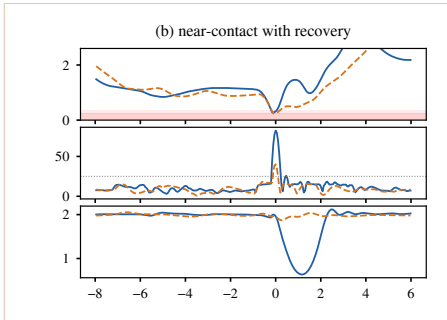


L-shape, PID-LQR · the closest drone-to-drone pass, 0.79 m · Click the frame to play

- Closest pair: 0.79 m (L-shape, PID-LQR)
- Per-run minima range up to 1.77 m
- The system tracks $d_{\min}$ only — it never detects a contact
- $\lambda_{ij} + \lambda_{ji} = 1$ keeps it symmetric: both give way

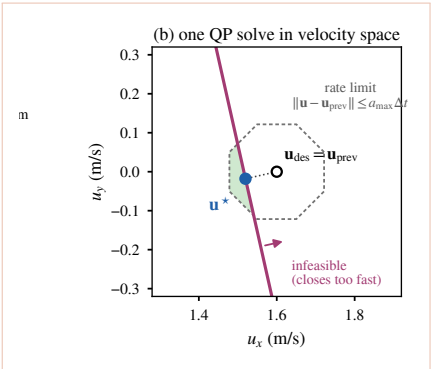
It can be neither confirmed nor excluded — and we say exactly that

- All 8 missions logged a negative barrier, to -0.68 m
- That implies about 0.34 m of surface distance
- Outside the 0.235 m body, inside the 0.348 m rotor reach
- Rotor guards would enlarge the envelope — but they are an assumption, and would not have saved the rect/LQR agent



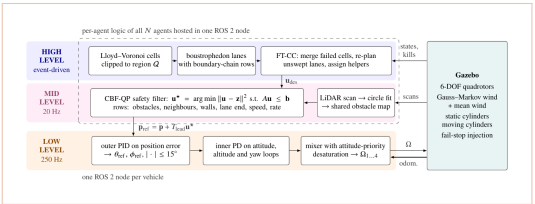
The filter uses about 0.3% of its own 50 ms budget

- One QP per agent per tick, exact active-set enumeration, ∼138 μs at 24 rows
- Pure numpy — no solver dependency, deliberately
- 19,500–25,900 solves per mission; tier 1–2 at 2.91–4.94%, tier 3 never in a completed run
- The 0.12 real-time factor is Gazebo rendering seven drones, not this code



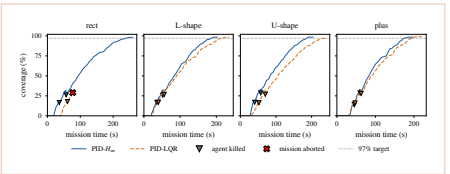
The filter scales; the upper layers are the real limit

- Rows grow with nearby hazards, not with fleet size — the QP scales
- But the reciprocal split is exact only from one snapshot, so one process holds all agents
- Partition, assignment and reallocation are centralised: one ROS 2 node
- Seven matched the arena, not a computational ceiling



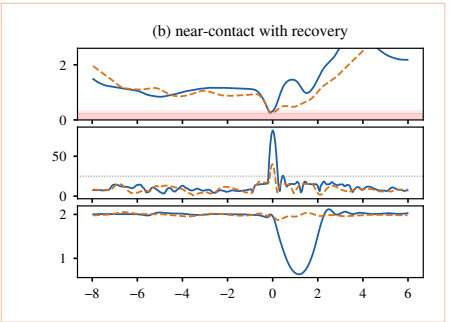
The flown coverage sits within a point of its own planner's ceiling

- A 0.30 m grid cell counts once a 0.95 m sensor disc passes above 0.8 m
- Offline upper bound for the planned lanes: 97.8–99.6%
- So the residual is cell geometry and lane spacing, not sweep failure
- The weakest completed run, U-shape / LQR at 96.9%, is reported rather than re-run



15° caps the attitude reference, not the physical attitude

- The cap applies to θ_{ref} leaving the outer loop
- During steady sweeping the 95th percentile is 14.2–14.8° — right at the cap
- Maxima come from the ~180° lane-end pivots and from upsets
- Extremes are diagnostic, not nominal: 155.9° is the crash, 82.5° the recovered graze



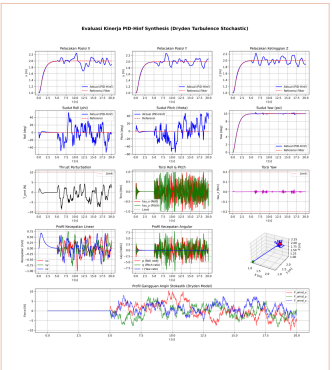
B1 · The acceleration budget behind $\varphi(h)$

- The a in the barrier is not $a_{\max}$: part of the envelope is reserved for the hazard's own motion and for wind
- $\mu = 0.75$ for obstacles, 0.85 for arena walls; wind allowance 0.5 m/s^2
- The floor keeps the barrier defined even when the budget is eaten

$$a = \max\left(0.15\, a_{\max},\ \mu\, a_{\max} - |a_{\text{obs}}| - |a_{\text{wind}}|\right)$$

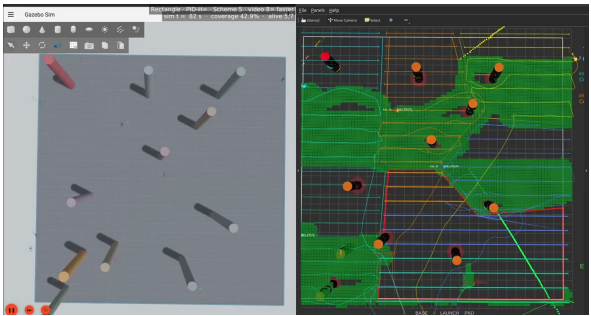
$$D(s) = \frac{s}{k_v}\ \ (s \le v_c), \qquad D(s) = \frac{s^2 + v_c^2}{2a_{\max}}\ \ (s > v_c), \qquad \varphi(h) = \gamma_r h\ \ (h < 0)$$

B2 · Offline step and turbulence response from gain synthesis



- Offline plant simulation used while choosing γ
- *Not* from the Gazebo campaign — shown only to illustrate the design step
- Every number in the talk comes from the flown runs instead

B3 · A cylinder found in flight re-plans only the lanes still ahead



rect, PID-H_∞ · 20 s around the closest static approach · Click the frame to play

- LiDAR confirms a cylinder never seen before
- Lanes below the sweep line are left alone
- Until the re-plan, the QP alone keeps it clear
- Closest static approach: 0.49 m

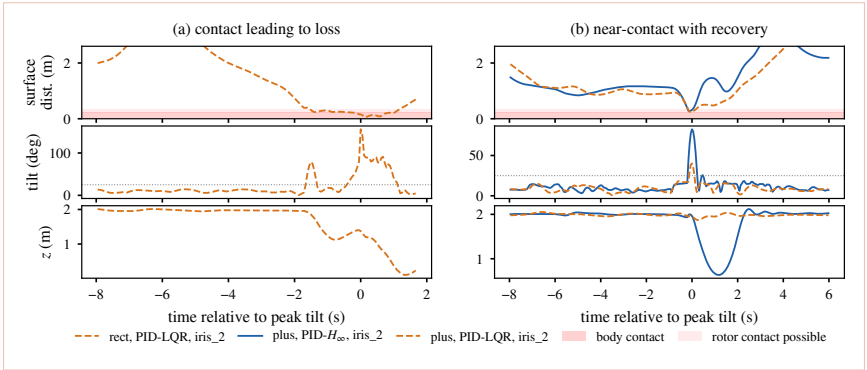
B4 · Gain synthesis and linear analysis in full

Design	Loop	K_p	K_i	K_d	$\|G\|_\infty$ (s^2)	ω_c (rad/s)	PM / GM
PID-LQR	outer	0.757	0.283	0.394	0.158	4.09	37.4° / 11.5 dB
	inner	4.868	–	0.534			
PID-H_∞	outer	2.705	1.092	1.107	0.040	9.08	25.2° / 9.0 dB
	inner	11.604	–	1.189			

Outer loops: $Q = \mathrm{diag}(0.08,\ 0.35)$, $R = 1$ for LQR; $\mathrm{diag}(0.10,\ 0.40)$ for H_∞.

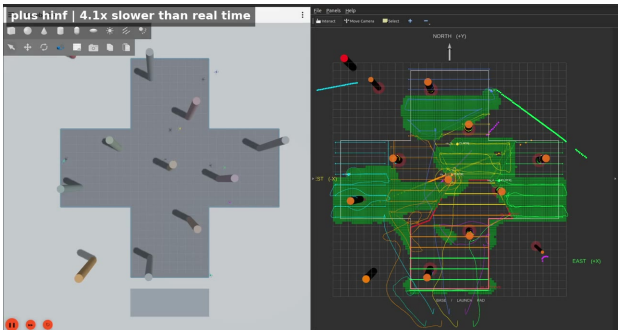
Inner loops: $Q = \mathrm{diag}(6,\ 0.6)$, $R = 0.05$; $\gamma_{\min} = 2.54$, used at $\gamma = 3.30$.

B5 · Upset events in full



(a) contact leading to loss — rect, PID-LQR, iris_2 · (b) near-contact with recovery — plus, both controllers, iris_2

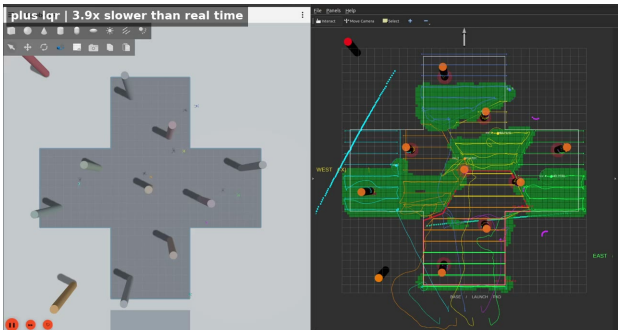
B6 · The H_∞ graze: a violent excursion that still recovered



Slowed 4.1× relative to real time · Click the frame to play

- Agent 2 grazes the central cylinder between lanes 4 and 5
- Tilt 82.5°, altitude to 0.64 m, clearance 0.28 m
- Recovered in about 3 s

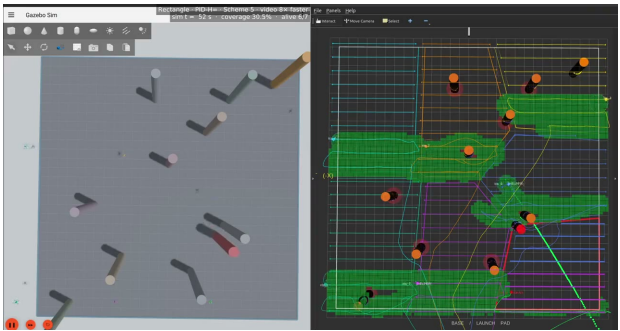
B7 · The same graze under LQR was milder



Slowed 3.9× relative to real time · Click the frame to play

- Peak tilt 40.1° against 82.5° for H_∞
- Altitude held at 1.87 m against 0.64 m
- The two are not ordered on every metric

B8 · The second loss needs three helpers



rect, PID-H_∞ · 25 s around the second kill · Click the frame to play

- Agent 7 killed at 32% coverage, agent 4 already gone
- The merged region is larger, so three helpers take it
- Helper groups across both losses: 7,6; 6,1,2

B9 · Older data in this repository disagrees, and here is why

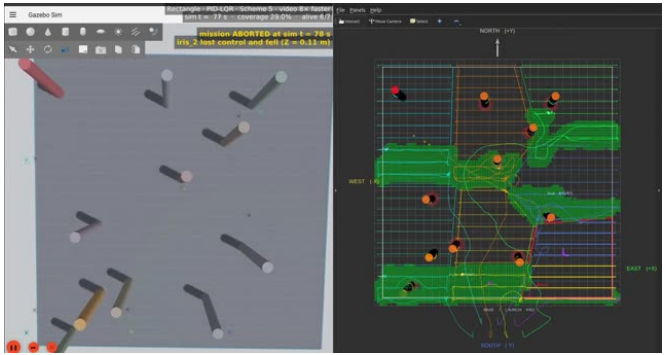
- A 3 September 2026 sweep under `results/skema5/` shows eight FAIL runs
- Superseded configuration; only 10 of its 40 matrix rows ever ran
- Four benchmark figures were deleted in August 2026 after the tooling was found to synthesise telemetry when CSVs were missing
- Current reporting prints n/a and never substitutes a value
- Every number in this talk comes from the 11–12 September campaign

B10 · All eight recordings are available here and published

- PID-H_∞ — rect 98.2% · L-shape 98.4% · U-shape 98.5% · plus 98.4%
- PID-LQR — rect aborted · L-shape 98.5% · U-shape 96.9% · plus 99.0%
- The next slides hold the full recordings — jump to one if a specific region comes up

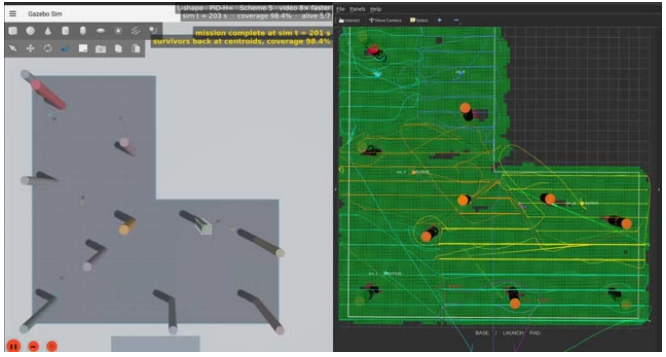
public playlist

B11 · Full mission — rect, PID-LQR (aborted)



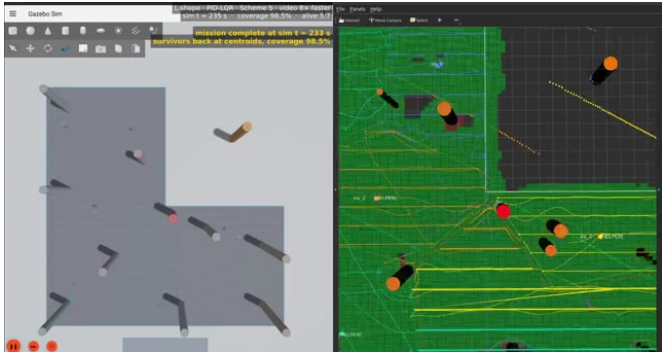
stopped at 78.8 s — the full recording of the failure · Click the frame to play

B12 · Full mission — L-shape, PID-H_∞



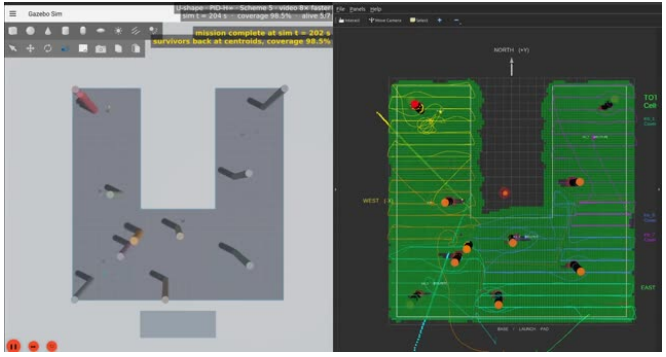
98.4% in 201 s · Click the frame to play

B13 · Full mission — L-shape, PID-LQR



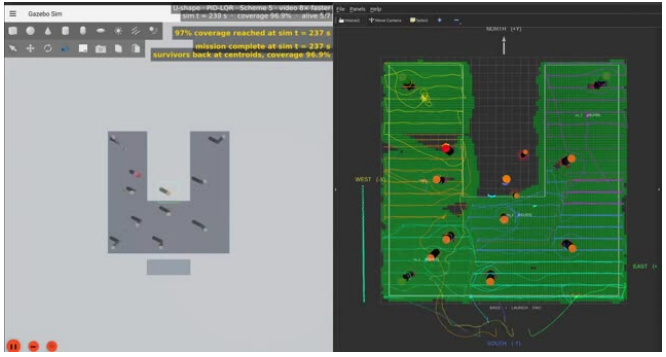
98.5% in 233 s · Click the frame to play

B14 · Full mission — U-shape, PID-H_∞



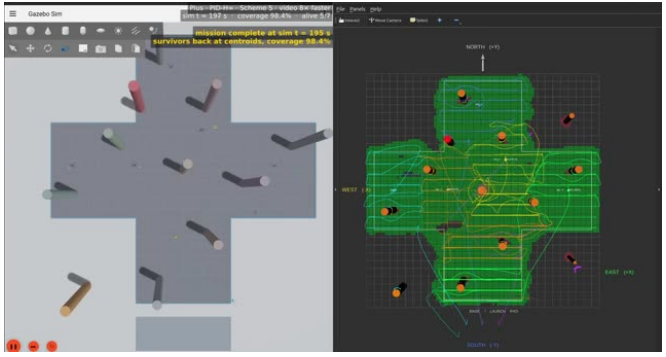
98.5% in 202 s · Click the frame to play

B15 · Full mission — U-shape, PID-LQR



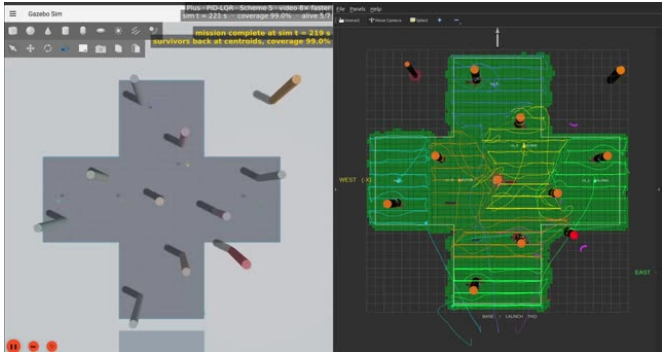
96.9% in 237 s · Click the frame to play

B16 · Full mission — plus, PID-H_∞



98.4% in 195 s · Click the frame to play

B17 · Full mission — plus, PID-LQR



99.0% in 219 s · Click the frame to play